

MUHAMMAD ALI NASIR

Java Backend Developer | Software Engineer

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PROFESSIONAL SUMMARY

Aspiring Java Backend Developer with hands-on experience developing and deploying enterprise web applications during a Software Engineering internship at Sui Southern Gas Company (SSGC). Experienced in building secure, scalable backend solutions using Java, Spring Boot, Spring Security, REST APIs, relational databases, and modern DevOps practices. Passionate about solving real-world business problems through clean architecture, API-driven development, and enterprise software engineering while continuously expanding expertise in distributed systems, cloud technologies, and AI-powered applications.

TECHNICAL SKILLS

Languages	Java 21, SQL, JavaScript
Backend Development	Spring Boot, Spring Security, Spring Data JPA, Hibernate, REST API Development, JWT Authentication, Role-Based Access Control (RBAC), Server-Side Pagination, Microservices, MVC Architecture
Databases	PostgreSQL, MySQL, Oracle Database, Redis
Messaging & Streaming	Apache Kafka
DevOps & Deployment	Docker, GitHub Actions (CI/CD), Maven, Git, VPS Deployment
API & Monitoring	Swagger / OpenAPI, Prometheus, Grafana
Frontend	Thymeleaf, HTML5, CSS3, Chart.js
AI & Integration	OpenAI API
Software Engineering	OOP, Data Structures & Algorithms, Database Design, Clean Architecture, SOLID Principles

PROFESSIONAL EXPERIENCE

Software Engineering Intern Aug 2025 – Sep 2025

Sui Southern Gas Company (SSGC) — Karachi, Pakistan

- Developed the Field Activity Reporting System (FARS), an internal web application that streamlined access to operational field activity data, reducing dependence on manually written SQL queries for recurring information requests.
- Collaborated with senior engineers and project stakeholders to design application features and integrate the platform with SSGC's production Oracle database.
- Built approximately 20–25 REST APIs using Java, Spring Boot, Spring Security, and Hibernate/JPA to support secure data retrieval and backend business operations.
- Implemented server-side pagination and optimized data retrieval to efficiently handle large datasets while improving application responsiveness.
- Developed hierarchical filtering capabilities across regions, units, and sub-units, enabling management to quickly retrieve operational information.
- Created interactive dashboards using Chart.js to visualize field activity data and support internal operational monitoring.
- Delivered the application for internal deployment at SSGC, where it is actively used by supervisors and management for field activity tracking and reporting.

PROJECTS

NexusERP Cloud (Ongoing) — Multi-Tenant Enterprise ERP SaaS Platform

Tech Stack: Java 21, Spring Boot 3, Spring Security, Spring Data JPA, Hibernate, PostgreSQL, Redis, Docker, GitHub Actions, Swagger/OpenAPI, JWT, RBAC

- Architected and developed a cloud-native multi-tenant ERP platform enabling multiple organizations to securely operate on shared infrastructure while maintaining complete tenant data isolation.
- Designed a comprehensive role-based access control (RBAC) system supporting organization administrators, HR, finance, inventory, procurement, sales, managers, and employees.

- Developed modular enterprise services for Organization Management, Inventory, Procurement, Sales, Human Resources, Finance, Employee Management, and Audit Logging using Spring Boot.
- Implemented tenant-aware JWT authentication and request filtering to enforce secure data segregation across organizations.
- Optimized application performance using Redis caching, efficient JPA query design, pagination, and database indexing.
- Containerized the platform with Docker and automated build workflows using GitHub Actions CI/CD.
- Documented REST APIs using Swagger/OpenAPI while following Clean Architecture, SOLID principles, and RESTful API best practices.
- Integrated AI-powered business insights to generate natural language reporting, sales analysis, inventory recommendations, and business performance summaries.

PipelineIQ — AI-Powered CI/CD Failure Analysis Platform

Tech Stack: Java 21, Spring Boot, Spring Security, Apache Kafka, PostgreSQL, Redis, Docker, GitHub Actions, Swagger/OpenAPI, OpenAI API, Prometheus, Grafana

- Designed and developed an event-driven platform that automates CI/CD failure analysis using Apache Kafka for asynchronous service communication.
- Integrated AI-powered log analysis to identify root causes, classify build failures, estimate severity, and recommend actionable resolutions.
- Built secure REST APIs with JWT authentication and role-based access control (RBAC) to manage application access.
- Implemented Kafka producers and consumers to enable reliable communication between distributed services.
- Automated build workflows using GitHub Actions and containerized application services with Docker.
- Configured Prometheus and Grafana dashboards to monitor service health, Kafka metrics, and CI/CD pipeline performance.
- Published interactive API documentation using Swagger/OpenAPI to simplify API testing and integration.

Abroad School Management System — School Administration & Student Management Platform

Tech Stack: Java, Spring Boot, Spring Security, MySQL, REST APIs, VPS Deployment

- Developed and deployed a web-based school management platform supporting admissions, fee management, student records, and staff administration.
- Designed backend services and REST APIs to manage core academic and administrative workflows.
- Implemented secure authentication and role-based access control to protect sensitive institutional data.
- Deployed the application on a VPS server for real-time operational use by the school.
- Optimized database operations and backend architecture to improve application reliability and maintainability.

EDUCATION

Bachelor of Science in Computer Science (BSCS) Jan 2023 – Jan 2027 (Expected)

Hamdard University, Karachi, Pakistan — CGPA: 3.10 / 4.00 — Completed 6th Semester, Entering 7th Semester

Higher Secondary School Certificate (Pre-Engineering) Jan 2021 – Jan 2023

Govt. Dehli College, Karachi, Pakistan

CERTIFICATIONS

Cloud Computing Certification — NED University of Engineering & Technology Sep 2025 – Nov 2025